MGT-8803-SC · NLP and Generative AI in Finance · Final Project

HELIOS

Agentic AI for Private Equity Take-Private Diligence

Compressing 4 to 8 weeks of diligence into 2 minutes...

AUTHORS / STUDENTS

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DATE

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REPOSITORY

github.com/anay-a-joshi/agentic-diligence

Executive Summary

An agentic AI platform that automates take-private screening for private equity firms.

PROBLEM

PE associates spend 4-8 weeks and $300K-$800K per take-private candidate, manually parsing SEC filings and building LBO models. Roughly 95% of screened candidates never become deals.

SOLUTION

HELIOS deploys 8 specialist AI agents that read SEC EDGAR filings, build a Bull/Base/Bear LBO model, and produce a downloadable IC memo and editable Excel model - in approximately 2 minutes.

MARKET

Approximately 5,000 active U.S. private equity firms manage over $2 trillion in dry powder. At a 5% penetration rate and $50K average seat price, the addressable market exceeds $250 million annually.

BUSINESS MODEL

Enterprise SaaS at $50K-$200K per analyst seat per year. Revenue streams include subscriptions, API metering, and white-label licensing. Target gross margin of 80%+ at steady state.

~2 min	8	$300K+
RUNTIME PER ANALYSIS	SPECIALIST AGENTS	COST COMPRESSION PER SCREEN

Industry Dynamics and the Pain Point

Why traditional take-private screening is structurally broken - and why the technology to fix it has just arrived.

Up to 4 weeks of analyst work → 2 minutes · approximately 1,440x acceleration

THE STATUS QUO

Manual Take-Private Screening

- 4-8 week diligence cycle per candidate
- $300K-$800K labor cost per single screen
- Associates working 80-100 hour weeks
- ~95% of screens never become deals
- SEC filings scattered across portals
- Reproducibility and audit trail are weak

WHY NOW

Three Converging Tailwinds

- Foundation LLMs (Llama 3.3 70B) reached production-grade reasoning in 2024
- SEC EDGAR XBRL API made financial data programmatically accessible
- Record $2T+ dry powder pressures faster screening cadence
- PE incumbents (Apollo, KKR, Blackstone) actively building internal AI teams

Solution Architecture: Eight Specialist Agents

A coordinated multi-agent pipeline reads SEC filings and produces investment-grade outputs.

1 Financial Agent

Extracts revenue, EBITDA, FCF from XBRL data; parses MD&A for drivers and risks.

2 Commercial Agent

Identifies operating segments, products, and named competitors from 10-K Item 1.

3 Risk Agent

Scores top five risk factors with severity classifications from 10-K Item 1A.

4 Governance Agent

Assesses board structure and takeover defenses from DEF 14A proxy statements.

5 Market Agent

Pulls live price, market capitalization, P/E, and beta from market data feeds.

6 Sentiment Agent

Analyzes management tone in MD&A; detects directional shifts year over year.

7 Red Flag Agent

Scans recent 8-K filings for material adverse events and governance changes.

8 Synthesis Agent

Consolidates the seven preceding outputs into an Investment Committee memo.

DELIVERABLES

0-100 Feasibility Score · 3-page IC Memo (PDF) · Live LBO Model (Excel) · Bull / Base / Bear Scenarios

TECH STACK

Llama 3.3 70B (Groq) · FastAPI · Next.js 15 · Tailwind · SEC EDGAR XBRL · yfinance · ReportLab · openpyxl

Working Prototype: Validated on Dell Technologies

End-to-end demonstration on a real take-private candidate, generating live SEC-grounded outputs.

LIVE DASHBOARD

Dell Technologies Inc.

DELL · Technology · CIK 1571996

40

/ 100 Grade F

Pass - Fundamental Issues

Recommendation: PASS

Base-case modeled returns of -14.6% IRR are below typical PE hurdle.

⚠ SIZE-PROHIBITIVE

$140.4B market cap exceeds $50B PE LBO ceiling.

LBO RETURNS · 5-YEAR HOLD

BULL CASE	MOIC 1.63x	+10.3%
BASE CASE	MOIC 0.45x	-14.6%
BEAR CASE	MOIC 1.09x	+1.8%

HEADLINE FINANCIALS · FY2026 · SOURCED FROM SEC EDGAR XBRL

REVENUE	EBITDA	FCF	MKT CAP
$88.4B	$8.7B	$5.9B	$140.4B

1,500+ lines Python backend · 2,000+ lines TypeScript frontend · 21+ Git commits · End-to-end working prototype on localhost

Market Opportunity and Business Model

A $250M+ addressable market in PE diligence, monetized through enterprise SaaS subscriptions.

MARKET SIZING

TAM

$250M+

5,000 PE firms × ~$50K/seat × 5% penetration

SAM

$120M

Mid-market and large PE firms in North America

SOM (Y3)

$5M ARR

10 enterprise logos at $500K average ACV

PRICING

- Tier 1 (Boutique): $50K/seat/yr
- Tier 2 (Mid-market): $100K/seat/yr
- Tier 3 (Enterprise): $200K/seat/yr
- Volume discount above 10 seats

REVENUE STREAMS

- Annual SaaS subscriptions (~80%)
- API metering for headless usage (~15%)
- White-label licensing (~5%)
- Premium support upsell

UNIT ECONOMICS

- CAC: ~$15K per logo
- LTV/CAC target: 5x at maturity
- Gross margin: ~80% at scale
- Compute: ~5% of revenue

Competitive Landscape and SWOT

HELIOS is the only agentic, output-grade screening platform; incumbents remain data providers.

COMPETITOR MATRIX

Capability	HELIOS	CapIQ	PB
Agentic AI pipeline	✓	—	—
Auto LBO model	✓	—	—
IC memo generation	✓	—	—
Live editable Excel	✓	—	—
Real-time SEC ingestion	✓	✓	✓
0-100 feasibility score	✓	—	—
Take-private screening	✓	—	◐

CapIQ = S&P Capital IQ; PB = Pitchbook; ◐ = partial.

SWOT ANALYSIS

STRENGTHS

- First-mover in agentic diligence
- 8-agent architecture vs single LLM
- Real deliverables: PDF + Live Excel
- Production-grade UI

WEAKNESSES

- Single foundation model dependency
- No proprietary data moat
- Requires PE workflow integration
- Early-stage brand

OPPORTUNITIES

- 5,000 US PE firms TAM
- Expansion to IB and corp dev
- International PE markets (EU, Asia)
- Adjacency: roll-up screening

THREATS

- Incumbents add agentic features
- Foundation model commoditization
- AI regulation in finance
- Talent compensation pressure

AI-Native Competitors and Defensible Moat

Beyond legacy data providers - how HELIOS holds the line against the next wave.

AI-NATIVE COMPETITOR LANDSCAPE

Competitor	Focus	Gap vs. HELIOS
AlphaSense	Search across filings & transcripts	No LBO model, no IC memo, no scoring
Hebbia	Document Q&A for finance	Generic; not PE-workflow native
Rogo	Investment banking workflows	IB-focused; no take-private screening
BlueFlame AI	Alt-asset GP workflow assistant	Workflow tool, not output generator
Internal PE AI	Apollo, KKR, Blackstone in-house	Not commercial; built for one firm

WHY HELIOS WINS

Verticalized for take-private screening - not a horizontal Q&A tool. Produces investment-grade deliverables (PDF memo + live Excel), not just answers. Other startups stop at retrieval; HELIOS goes through to recommendation.

DEFENSIBLE MOAT (4 LAYERS)

1. **Orchestration layer -** Proprietary 8-agent coordination logic; not the LLM itself
2. **Domain prompts -** Structured prompts encoding PE diligence heuristics
3. **Scoring algorithm -** 6-dimension weighted model refined via customer feedback
4. **Workflow lock-in -** IC memo + Excel become analyst's working artifacts

IP STRATEGY

Patents pending: multi-agent orchestration + feasibility scoring. Trade-secret on prompts/weights. Trademark on HELIOS mark.

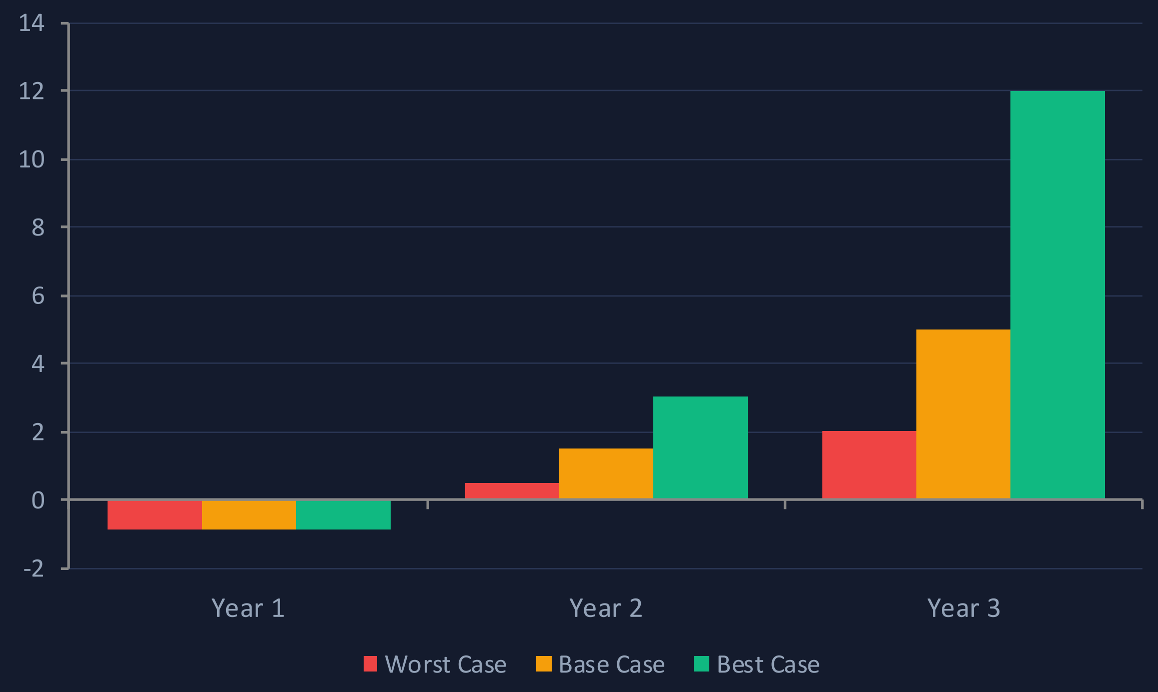
CAC vs. INCUMBENTS

HELIOS: ~$15K/logo (founder-led) · **Incumbents:** $50K-$100K (enterprise sales)

Financial Projections, Regulation, and Path to Exit

Three-year financial trajectory, regulatory positioning, and acquirer landscape.

3-YEAR ARR PROJECTIONS ($M, BY SCENARIO)



Startup costs: ~$850K Y1 burn · Cash flow positive Y2 · $5M ARR Y3 (base)

REGULATORY

- Not investment advice (clear disclaimer on every output)
- SEC EDGAR data is public domain - no licensing fees
- SOC 2 Type II planned for Year 2 enterprise sales
- Aligns with SEC Marketing Rule 206(4)-1 for advisor use

KEY RISKS

- Foundation model regression on prompt sensitivity
- SEC API changes (mitigated via 24-hour result cache)
- Incumbent response from CapIQ / Pitchbook
- Compute cost shocks if Groq pricing changes

EXIT STRATEGY

- Strategic acquirer: S&P Global, Moody's, Bloomberg
- Adjacent acquirer: Salesforce, Oracle, SAP
- Indicative valuation: 12-18x ARR (SaaS comps)
- Target horizon: Year 4 to Year 6

Funding Ask and 18-Month Roadmap

$1.5M seed gets HELIOS to revenue and a defensible market position.

THE ASK

$1.5M
SEED ROUND

18 mo
RUNWAY TO BREAKEVEN

$6M
PRE-MONEY VALUATION

~20%
EQUITY OFFERED

USE OF FUNDS

60%	Engineering (2 senior + 1 ML)
25%	GTM (founder-led + 1 BD)
10%	Infra & compute
5%	Legal & SOC 2

TARGET INVESTORS

- Bessemer Venture Partners
- Nyca Partners
- QED Investors
- Strategic angels: Apollo / KKR / Blackstone alumni

18-MONTH ROADMAP

Q3'26	Seed close · 3 design partners
Q4'26	SOC 2 Type I · multi-LLM
Q1'27	First 3 paid logos
Q2'27	$500K ARR · Series A talks
Q3'27	10 logos · $1M ARR · Series A

Team and Recommendation

The team, the verdict, and what we're asking the room to do next.

THE TEAM

AAJ — **Anay Abhijit Joshi**
Lead Author & Technical Architect
Georgia Tech MS-QCF
Full-stack engineering, agent orchestration, LBO modeling

VC — **Vidhi Chaturvedi**
Product & Market Strategy
Georgia Tech MS-QCF
Market sizing, go-to-market, business model design

HB — **Himanshu Bhatt**
Financial Modeling & Validation
Georgia Tech MS-QCF
LBO engine validation, financial analysis, PE workflow

EXECUTION-RISK ASSESSMENT

Foundation-model dependency	→	Multi-LLM redundancy roadmapped Q4 2026
SEC API drift	→	24-hour result cache; nightly schema validation
Incumbent fast-follow	→	Patent filings; workflow lock-in via deliverables
PE sales-cycle length	→	Design-partner program; founder-led warm intros

RECOMMENDATION

PROCEED

Working prototype · validated economics · defensible orchestration moat · clear path to liquidity at 12-18x ARR.

THE ASK TO THIS ROOM: Intros to PE design partners · Seed investor introductions · Feedback on prompt library and scoring weights

Thank You

Questions & Discussion

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